HUAWEI | IRELAND RESEARCH CENTER, SNI Lab



Faster Vector + Tag Retrieval in Lance

Predicate-result cache · Filter-aware partition skipping · Batched mining

Research proposal v1 · 11 November 2026

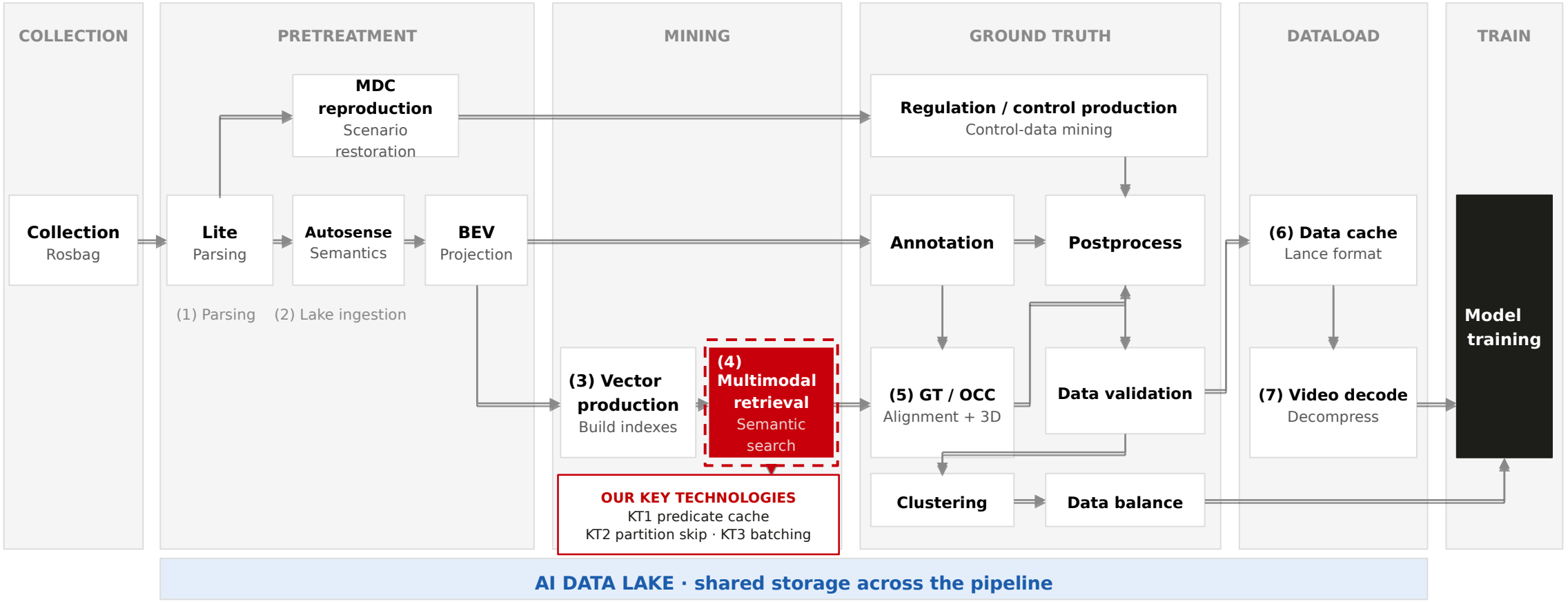
Vector Index Optimization

Huawei Technologies Co., Ltd.

HUAWEI

We speed up ADS stage (4): vector + tag retrieval

[Company context] Redrawn from ADS-PLAN slide 1; target is (4) Multimodal retrieval: vector search joined with scalar tags.



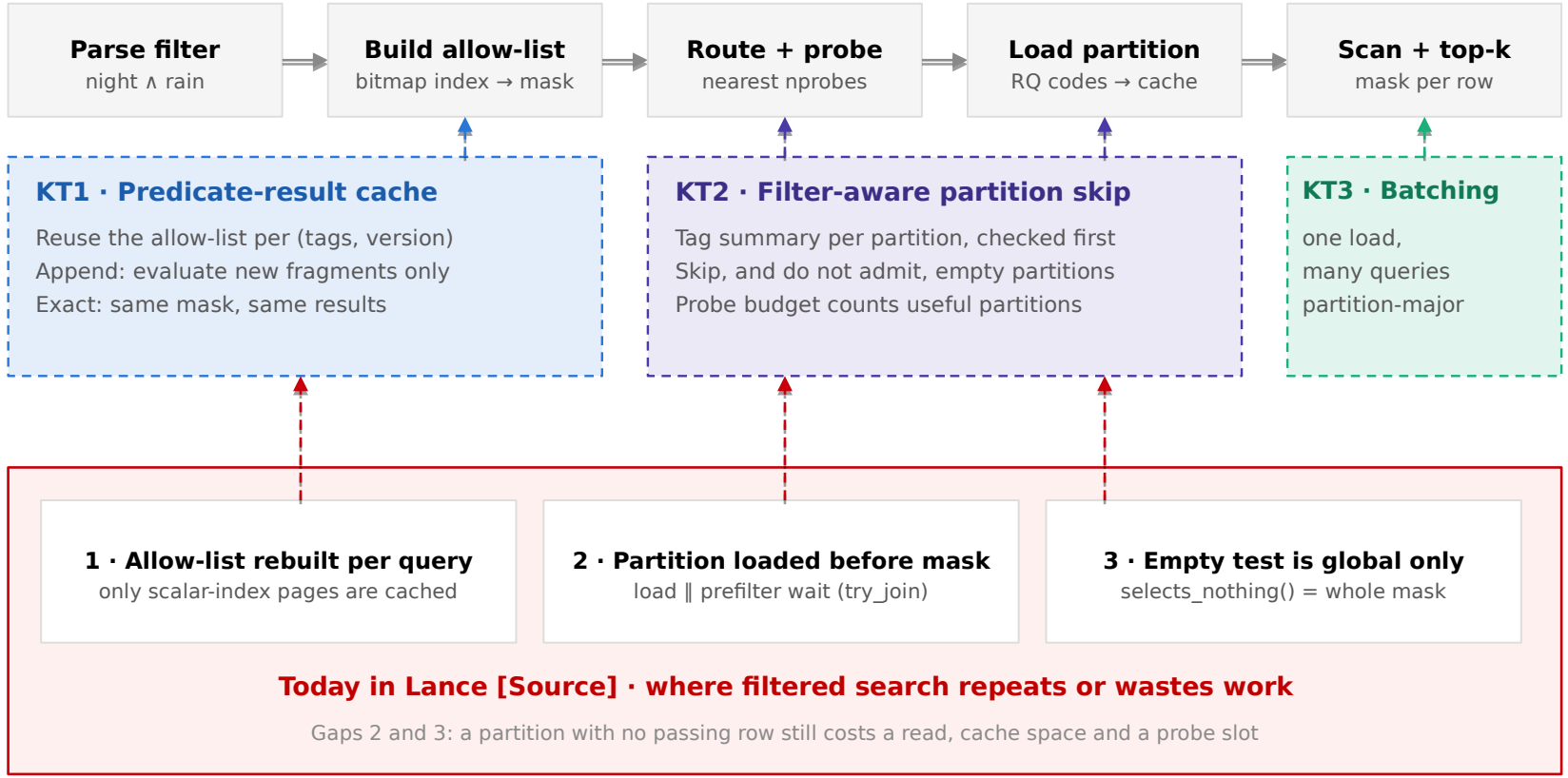
Mining runs many filtered vector searches over one lake: work repeats. All three key technologies sit in this box.

– Huawei Confidential –

Three key technologies in Lance filtered search

Lance IVF_RQ query with a tag prefilter, each key technology (KT) at its integration point · gaps [Source] · dashed = proposed

LANCE FILTERED VECTOR SEARCH (IVF_RQ + TAG PREFILTER)



STATUS · 11 NOV 2026

KT1 predicate cache

- Gap verified in Lance source
- Design after SIGMOD 2024 paper

KT2 partition skip

- Gap verified: load before mask
- Model: 54% of probes empty
- at 0.1% tags, K = 16,384

KT3 batched mining

- Reported 4–8× (SieveIVF, on Lance)
- ADS trace not yet collected

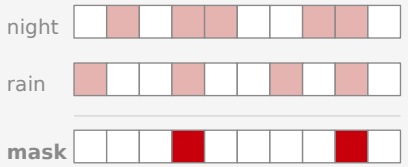
KT1 removes repeated filter work; KT2 and KT3 remove wasted partition loads. Each ships separately.

– Huawei Confidential –

Filtered IVF loads partitions where no row passes

One filtered query in Lance today, redrawn by us from the source [Source] · red tags = the key technology that removes the cost

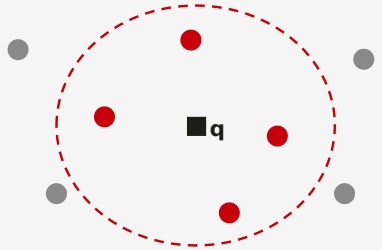
1 · Evaluate tag filter



Bitmaps are ANDed into an allow-list; pages are cached, the result is not

REPEATED PER QUERY → KT1

2 · Rank partitions



nprobes nearest centroids; the filter is not consulted

BLIND TO FILTER → KT2

3 · Load + admit

P7 · 3 pass

P2 · 0 pass loaded anyway

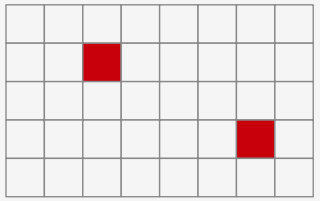
P9 · 1 pass

P4 · 0 pass loaded anyway

Runs in parallel with the mask: read and cached even if no row passes

WASTED I/O → KT2

4 · Scan with mask



score all · keep 2

Every row is scored, then rows outside the mask are dropped

WASTED COMPUTE → KT2

Model · share of probed partitions with no passing row

Tag selectivity s	0.01%	0.1%	0.5%	1%
K = 16,384 (≈ 610 rows / partition)	94%	54%	4.7%	0.2%
K = 4,096 (≈ 2,441 rows / partition)	78%	8.7%	0.0%	0.0%

10M rows, tags independent of the embedding: P(empty) ≈ e^(−s·n). Measure on ADS.

Why ADS mining is the bad case [Assumption]

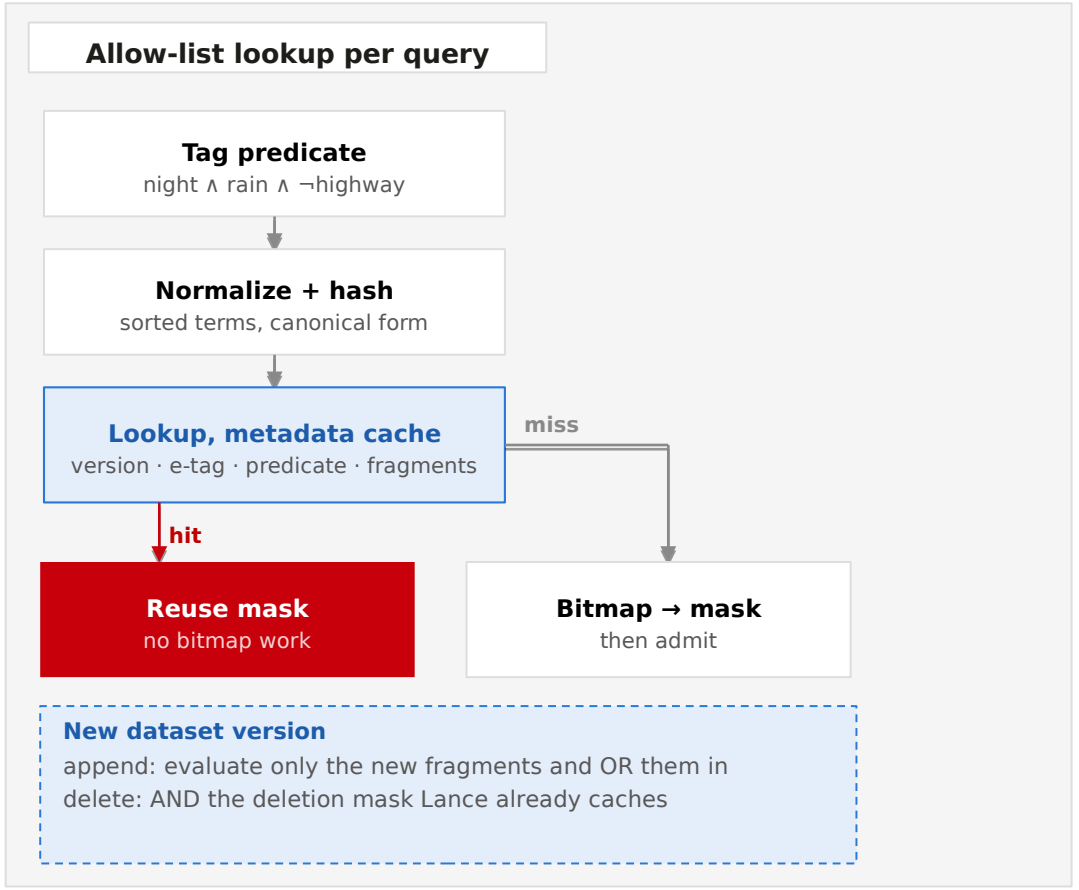
· rare scenarios combine several tags → selectivity well below 1%
· the same tag sets are reused with many query vectors
· batch jobs: throughput counts more than one query's p99

At selective filters, most probed partitions return nothing but are still loaded; the filter result is rebuilt every time.

– Huawei Confidential –

KT1 · Cache the tag allow-list, not only pages

[Proposal] after Predicate Caching (Schmidt et al., SIGMOD 2024) · key and invalidation follow Lance's existing RowAddrMaskKey [Source]



Design [Proposal] on existing Lance pieces [Source]

Aspect	Design	Lance piece
Key	version · e-tag · predicate · fragments	RowAddrMaskKey pattern
Value	RowAddrMask (roaring allow-list)	DatasetPreFilter.filtered_ids
Append	new fragments only, OR into mask	fragments are immutable
Delete	AND the cached deletion mask	create_deletion_mask
Admission	eval time × reuse ÷ mask bytes	metadata cache (S3-FIFO)

Evidence and what we measure first

Item	Value
Predicate Caching in Redshift [Reported]	up to 10× on selected queries
Share of query time spent building the mask, ADS	**TBD**
Predicate repeat rate in the ADS mining trace	**TBD**

> **First gate:** repeat rate and mask-time share on the ADS trace justify a cache; results stay identical.

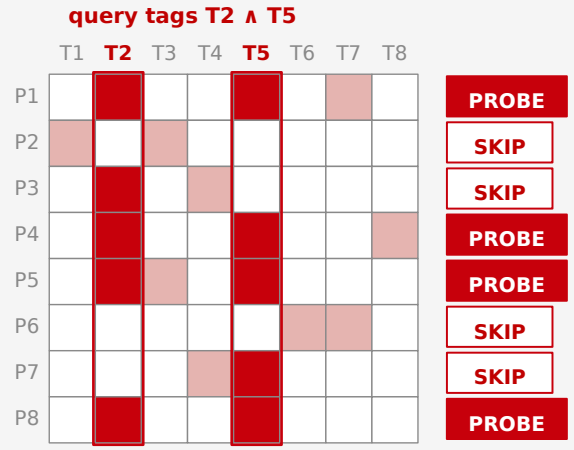
Same tags, same version, same mask: compute it once and reuse it exactly.

– Huawei Confidential –

KT2 · Skip partitions where no row can pass

[Proposal] conservative tag summary per IVF partition, checked before load · adaptive stop as in SieveIVF (on Lance) and VADER (VLDB 2027)

Partition × tag summary (K × T bits)



by distance

· Built at index build / optimize: 16,384 × 1,024 tags = 2 MiB [Model]
· Conservative: a partition is skipped only if a required tag is absent
· Never skips a partition with a passing row; stale after deletes is still safe
· NOT / range predicates the summary cannot decide → probe as today

Change in prepare_partition

Today	**load ‖ wait mask** read + admit always	→ **scan with mask** 0-row: scored, dropped
KT2	**summary check** skip: no read, no admit	→ **load → scan** useful partitions only

Expected effect [Model] and prior art [Reported]

Item	Value
Probe reads avoidable, 0.1% tags, K = 16,384 [Model]	**≈ 54%**
SieveIVF on Lance, 10M / 100M rows [Reported]	4.1–7.6× / 6.1–8.4×
SieveIVF filtered recall loss [Reported]	0.03–2.29 pp
VADER vs best baseline [Reported]	up to 53% faster
Lakehouse: files pruned before ANN [Reported]	355 of 444

First gate: recall@10 equal to today at the same useful-probe count, and partition reads per query fall as the model predicts on ADS tags.

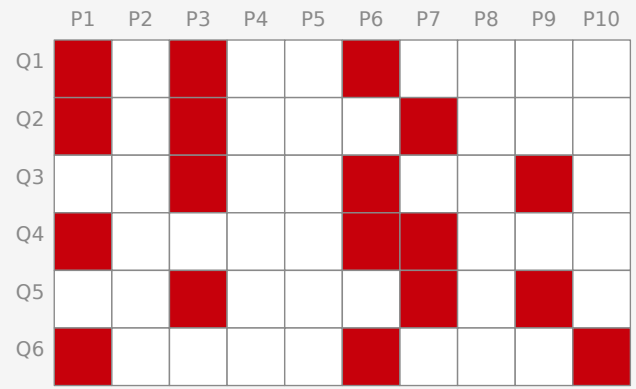
Check the summary before the read: empty partitions cost nothing and free their probe slot.

– Huawei Confidential –

KT3 · Batch mining queries that share partitions

[Proposal] partition-major execution over a window of queries · after SieveIVF continuous batching, CaGR-RAG and CALL query grouping

Queries × probed partitions (after KT2 skip)



18
loads, query by query

6
loads, partition-major

Illustrative window of 6 queries: 18 partition reads become 6; each loaded partition is scanned once for every query that probes it, with each query's own mask (KT1).

How a batch runs [Proposal]

- collect a window of (vector, tags) mining queries
- route all queries, skip empty partitions (KT2), union the rest
- load each partition once, scan it for every query that needs it
- batch loads use their own admission class: no online-cache pollution

Prior art [Reported]

System	Idea	Reported
SieveIVF (on Lance)	group ready queries by partition	part of 4.1–8.4×
CaGR-RAG	group queries by shared clusters	p99 up to −51.55%
CALL	context-aware grouping	p99 up to −33%

First gate: mining throughput up at equal recall@10, and online query p99 unchanged while a batch runs.

Mining is batch work: read a shared partition once, not once per query.

– Huawei Confidential –

Summary · recall-safe by design, ADS trace next

Three key technologies · gaps verified in Lance source · [Proposal] sizes S ≈ 1–2, M ≈ 3–5, L ≈ 6–10 engineer-weeks

KT1 · Predicate cache

TBD

mask time saved on ADS

· reported up to 10× (Redshift)
· exact: identical results

KT2 · Partition skip

≈ 54%

probe reads avoidable
[Model]

· at 0.1% tags, K = 16,384
· reported 4–8× (SieveIVF)

KT3 · Batched mining

TBD

mining throughput on ADS

· reported p99 −33 to −52%
· one read, many queries

ROADMAP · EACH PHASE HAS A GATE

1 · ADS trace + baseline · S → **2 · KT2 partition skip · M** → **3 · KT1 predicate cache · S** → **4 · KT3 batching, upstream · M**

tag selectivity, repeat rate, reads per query

tag summary, check before load

allow-list in metadata cache

batched search, Lance PRs

Recall-safe by default: KT1 is exact, KT2's summary never skips a partition with a passing row, adaptive stopping is opt-in; a failed gate keeps today's path.

The gaps are in the code today; ADS data decides the size of the gain. Phase 1 fills every TBD.

– Huawei Confidential –